11/01/25 M/S

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FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

FUNDAMENTALS OF CIVIL ENGINEERING AND MATERIALS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Comment on scope of Civil Engineering	2	01	**02**
	b.	Describe the significance of pavements	2	02	**02**
	c.	Classify the types of paints based on application	2	03	**02**
	d.	Describe the purpose of lintels	2	04	**02**
	e.	Define the term "sustainability" in the context of modern development	2	05	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Describe the different types of civil engineering projects?	2	01	**05**
		OR			
	b.	Outline the challenges faced by civil engineers today	2	01	**05**
3.	**a.**	Describe the different types of roads based on function	2	02	**05**
		OR			
	b.	Explain the basic components of a bridge	2	02	**05**
4.	**a.**	Illustrate the requirements of a good stone	3	03	**05**
		OR			
	b.	Comment on the term concrete, explain its key components	3	03	**05**
5.	**a.**	Illustrate the requirements of a good brick	3	03	**05**
		OR			
	b.	Comment on characteristics that make it a valuable construction material	3	03	**05**
6.	**a.**	Explain the following terms: joint, bed, header, stretcher and quoin	3	04	**05**
		OR			

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	b.	Sketch and explain Rubble masonry and Ashlar masonry	3	04	**05**
7.	**a.**	Illustrate the necessity of Comprehensive development plan	3	05	**05**
		OR			
	b.	Comment on different types of buildings	3	05	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Write short notes on: Geotechnical, Environmental Engineering, Transportation Engineering and Concrete Technology	2	01	**10**
		OR			
	b.	Explain the role of Civil Engineers in Nation building	2	01	**10**
9.	**a.**	Explain any five types of dams in detail	2	02	**10**
		OR			
	b.	Explain the different types of infrastructure in Civil Engineering aspect	2	02	**10**
10.	**a.**	Illustrate the difference between fine and coarse aggregate in detail	3	03	**10**
		OR			
	b.	Illustrate the different types of cement	3	03	**10**
11.	**a.**	Explain the different types of floors	3	04	**10**
		OR			
	b.	Sketch and differentiate between lintel and arch	3	04	**10**
12.	**a.**	Comment on the materials used for doors and windows	3	04	**10**
		OR			
	b.	Sketch and explain the components of a good roof	3	04	**10**
13.	**a.**	Illustrate the different materials used for green building construction	3	05	**10**
		OR			
	b.	Comment on the factors considered in building planning	3	05	**10**